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Anatomical and Physiological Definition of the Motor Cortex of the Marmoset Monkey

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ABSTRACT

We used a combination of anatomical and physiological techniques to define the primary motor cortex (M1) of the marmoset monkey and its relationship to adjacent cortical fields. Area M1, defined as a region containing a representation of the entire body and showing the highest excitability to intracortical microstimulation, is architecturally heterogeneous: it encompasses both the caudal part of the densely myelinated "gigantopyramidal" cortex (field 4) and a lateral region, corresponding to the face representation, which is less myelinated and has smaller layer 5 pyramidal cells (field 4c). Rostral to M1 is a field that is strongly reminiscent of field 4 in terms of cyto- and myeloarchitecture but that in the marmoset is poorly responsive to microstimulation. Anatomical tracing experiments revealed that this rostral field is interconnected with visual areas of the posterior parietal cortex, whereas M1 itself has no such connections. For these reasons, we considered this field to be best described as part of the dorsal premotor cortex and adopted the designation 6Dc. Histological criteria were used to define other fields adjacent to M1, including medial and ventral subdivisions of the premotor cortex (fields 6M and 6V) and the rostral somatosensory field (area 3a), as well as a rostral subdivision of the dorsal premotor area (field 6Dr). These results suggest a basic plan underlying the histological organization of the caudal frontal cortex in different simian species, which has been elaborated during the evolution of larger species of primate by creation of further morphological and functional subdivisions. J. Comp. Neurol. 506:860–876, 2008. © 2007 Wiley-Liss, Inc.

Indexing terms: frontal lobe; somatotopic organization; cytoarchitecture; area 4; premotor cortex; New World monkey

This study forms part of an ongoing project aimed at understanding the organization of cortical areas in the marmoset (*Callithrix jacchus*), a species of New World monkey that has become an important model for the study of neurological and neuropsychological conditions (for recent examples, see Eslamboli et al., 2005; Brevard et al., 2006; Walker et al., 2006; Blezer et al., 2007). Here we use a combination of histological analyses, electrophysiology, and anatomical tracing to clarify the extent of the primary motor cortex (M1) relative to adjacent fields.

The relationship between the stimulation of sites in M1 and the excitation of specific muscle groups is consistent among primates: facial movements are represented most laterally, followed by those of the forelimb, trunk, and hindlimb/tail as the midline is approached (e.g., Penfield and Boldrey, 1937; Woolsey et al., 1952; Strick and Preston, 1978; Stepniewska et al., 1993). The presence of a

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complete skeletomotor representation, with movements elicited by relatively low currents in comparison with adjacent cortex, is therefore a key criterion for the physiological definition of M1. At the same time, the primary motor cortex has been traditionally equated with a histological field (Brodmann's area 4), which is characterized by features such as low cellular density, poor lamination, absence of layer 4, and presence of very large pyramidal cells in layer 5 (Betz, 1874; Lewis, 1878; Campbell, 1905; Brodmann, 1909; see Geyer et al., 2000, for review). However, studies in Old World monkeys have acknowledged that the histological character of M1 may change, with layer 5 pyramidal cells becoming smaller from medial to lateral portions of this area (e.g., Vogt and Vogt, 1919; Matelli et al., 1985). For example, the primary motor cortex of baboons has been shown to include three architectural subdivisions in mediolateral succession (4a, 4b, and 4c), which can be distinguished not only by the size of layer 5 pyramidal cells but also by the presence/absence of an incipient layer 4 (Watanabe-Sawaguchi et al., 1991).

The complexities involved in defining M1 have been further highlighted by studies in the New World monkeys, which added another dimension to the puzzle by describing caudal (M1c) and rostral (M1r) histological subdivisions in M1 (Stepniewska et al., 1993). These authors reported that "M1r" and "M1c" differ not only by the size of layer 5 neurons but also in terms of electrical excitability and connections. An analogous rostrocaudal subdivision of M1 has since been reported also in the macaque (Preuss et al., 1997). Given these caveats, one may question to what extent the physiologically defined M1 adheres to the traditional expectation of a cortical area being a physiologically, histologically, and connectionally well-defined "tile" of the cortical mosaic. Indeed, recent studies have questioned the validity of these expectations, with reference to other regions of the neocortex (cf. Rosa and Tweedale, 2005; Palmer and Rosa, 2006b; Graziano and Aflalo, 2007). The origin of corticospinal axons adds little light to the issue of defining M1, given that cells forming these projections are known to originate in many areas, including premotor and parietal cortices (Dum and Strick, 1991; He et al., 1993; Galea and Darian-Smith, 1994; Luppino et al., 1994; Galea, 1997; Wu et al., 2000).

We explored the relationship between histological fields and the somatotopic organization of electrically elicited movements and then compared the extent of the putative M1 with the sources of frontal projections to anterior and posterior parietal fields (regions that have major roles in somatic sensation and vision, respectively). Taken together, these results clarify the organization of the caudal frontal cortex of the marmoset and the homologies between fields in New and Old World monkeys.

MATERIALS AND METHODS

Experiments were approved by the Monash University Animal Experimentation Ethics Committee, which also monitored the welfare of the animals. Guidelines of the *Australian Code of Practice for the Care and Use of Animals for Scientific Purposes* were followed.

Microstimulation experiments

Intracortical microstimulation experiments were performed on one female and two male adult marmosets (*Callithrix jacchus*), with the objective of defining the representation of different body parts in M1. The animals were premedicated with intramuscular (i.m.) injections of diazepam (3.0 mg/kg) and atropine (0.2 mg/kg) and, after 30 minutes, anesthetized with ketamine (50 mg/kg), which provided deep anesthesia for the duration of the surgical procedures (>2 hours). A tracheotomy was performed, and a cannula was inserted into the saphenous vein of the leg ipsilateral to the cortical hemisphere under study, in order to allow a continuous infusion of a solution consisting of sufentanil (6 μg/kg/h) and dexamethasone (0.4 mg/kg/h) in a saline/glucose solution. The animal was placed in a stereotaxic frame and artificially ventilated with a mixture of oxygen and nitrous oxide (2:3). This anesthetic protocol was chosen as the one most compatible with microstimulation experiments in this species, after attempts using halothane, ketamine (i.m.), and telazol (i.m.), all of which produced disappointing results. Whenever movements were elicited, results from these early experiments revealed a topographic organization similar to the one observed in the cases reported here; however, these observations were not included in the present paper.

A craniotomy was performed and the dura mater folded back to expose the frontal cortex. The cortical surface was covered with warm agar to avoid desiccation. Using stereotaxic coordinates obtained in the course of previous studies of the marmoset frontal cortex (Burman et al., 2006), the cortex in the likely location of cytoarchitectural fields 3, 4, and 6 was stimulated, with penetrations forming rostrocaudal or mediolateral rows while avoiding surface blood vessels. Low-resistance (<1 MΩ) parylene-coated tungsten microelectrodes were used, penetrating to depths of up to 2,400 μm along the vertical stereotaxic axis. Given that in early experiments we found that the lowest thresholds were typically obtained at a depth of 2,100 μm, this depth was used for microstimulation in most penetrations (because the vertical plane is not perpendicular to the cortical layers in this part of the marmoset cortex, this depth corresponded to the cortical infragranular layers).

The microstimulation currents were delivered in 60-msec trains, with a pulse duration of 0.2 msec and a pulse frequency of 300 Hz. Penetration sites were stimulated initially with currents ranging from 50 to 200 μA in order to locate the topography of muscle representations, although at unresponsive sites higher currents (of up to 300 μA) were used. Once a motor response was elicited, the current was gradually reduced to identify the threshold current required to create a movement. Movements evoked by microstimulation were identified by at least two observers and recorded in digital video.

Tracer injections

In three animals we examined the relationship between histological subdivisions of the frontal cortex and the location of neurons with projections to the posterior parietal cortex. One of these animals (CJ45; see Fig. 8) was also part of the study of Burman et al. (2006); however, the present injections were not reported in the earlier study and were located in the opposite cerebral hemisphere. These animals were premedicated as described above and kept under ketamine anesthesia for the duration of the procedure (50 mg/kg initial dose, 10 mg/kg additional doses as needed). They were placed in a stereotaxic frame, and the parietal cortex was exposed. The retrograde fluorescent tracers diamidino yellow (DY) and fast blue (FB)

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were directly applied into the cortex as crystals (approximately 200 μm in diameter) with the aid of tungsten wires; Fluoro Ruby (FR) and Fluoro Emerald (FE) were injected (0.25 μl) via 1-μl syringes (Scientific Glass Engineering, Melbourne, Australia). The placement of the tracers was guided by stereotaxic coordinates obtained in the course of previous studies of marmoset visual cortex (Rosa and Schmid, 1995; Rosa et al., 2005), and their exact location in relation to cortical layers and areal boundaries was later assessed by histological reconstruction (e.g., Palmer and Rosa, 2006a,b). The cortex was covered with saline-soaked ophthalmic gelfilm, the piece of bone removed during the craniotomy was fixed back in place with dental acrylic, and the wound was closed in anatomical layers. Analgesics (oral paracetamol, 3 drops every 6 hours) and antibiotics (Norocillin, Norbrook, Newry, Northern Ireland; 0.1 ml) were given routinely for the first 24 hours post surgery. A survival time of 2 weeks was allowed.

Perfusion, fixation, and tissue processing

At the end of the microstimulation experiments, or after the survival time for retrograde transport of the tracers, the animals were overdosed with sodium pentobarbitone (100 mg/kg, i.v.) and perfused transcardially with 1 liter of heparinized saline followed by 1 liter of 4% paraformaldehyde in 0.1 M phosphate buffer (pH 7.4). The brains were postfixed in the same medium for a minimum of 24 hours before being cryoprotected by immersion in buffered paraformaldehyde solution containing increasing concentrations of sucrose (10%, 20%, 30%). Frozen 40-μm-thick sections were cut parasagittally or coronally using a cryostat. One series of sections was mounted onto gelatinized slides and air-dried for later staining with cresyl violet, whereas a second series was kept in phosphate buffer and stained free floating for cytochrome oxidase activity (Wong-Riley, 1979). A third series was stored in 10% formalin for at least 2 weeks before staining for myelin, with either the Gallyas (1979) stain or the gold chloride stain (Schmued, 1990). In the anatomical experiments a final series of sections was mounted unstained on gelatinized slides for the analysis of cells labeled with fluorescent tracers, after quick dehydration in 100% ethanol, clearing in xylene, and coverslipping with DPX.

Documentation of results

Dorsal reconstructions of the frontal cortex were prepared by aligning magnified drawings of sections 200 μm apart, using fiducial holes as references (created by introducing tungsten rods in the brain, prior to freezing and sectioning), and then graphically stacking projections of the dorsal surface onto a horizontal plane. The exact location of each microstimulation site was deduced by histological reconstruction of the electrode tracks relative to landmarks such as dimples on the cortical surface produced by large blood vessels (which were compared with photographs of the pattern of blood vessels obtained during the microstimulation experiments). Histological borders (see Results) were plotted as transition zones of various widths, which reflect sources of uncertainty such as the separation between sections used in the analysis, the test-retest variability (assessed by repeated plotting by the same observer on different days), the slightly different estimates provided by different architectural methods, and the interference of histology artifacts. Photomicrographs were obtained through Zeiss AxioCam and AxioVision v4.2 software (Carl Zeiss, Oberkochen, Germany). Low-power views typically involved the merging of multiple images, from adjacent portions of a same section, using Adobe Photoshop 8.0 (Adobe Systems, San Jose, CA). In such cases, the contrast and exposure of adjacent shots was matched by using standard software features, but no other adjustments were made. For lettering and figure composition, the images were imported into Canvas X (ACD Systems, Victoria, BC, Canada). Two-dimensional reconstructions of the cortical surface were also obtained by graphically "unfolding" contours of layer 4 of sections 400 μm apart, in such a way as to keep the neighborhood relationships between and within sections (Van Essen and Maunsell, 1980).

RESULTS

We will start the description of the results by illustrating the architectural characteristics of the cortex within and around M1, followed by a correlation between architectural fields and the somatotopic maps derived from intracortical microstimulation. We will then demonstrate the results of anatomical tracing experiments revealing interconnections between the agranular frontal areas and subdivisions of the posterior parietal cortex, which address the issue of the distinction between primary motor and premotor areas.

Architecture of the posterior frontal cortex

Architectural definition of the posterior and ventral borders of M1. In agreement with the primate norm, the marmoset frontal cortex reveals a caudal to rostral succession of motor, premotor, and granular frontal (prefrontal) fields, characterized by reduction in the size of pyramidal cells in layer 5 and increasing definition of layer 4 (Peden and Von Bonin, 1947; Burman et al., 2006).

Figure 1 illustrates parasagittal sections approximately 5 mm from the medial surface. (As indicated by the microstimulation results below, this level typically corresponds to the representation of the forelimb in M1.) Caudal to the frontal cortex are somatosensory fields 3b and 3a, which form successive narrow bands elongated in the posteromedial to anterolateral direction (Huffman and Krubitzer, 2001; Burman et al., 2006). In Nissl-stained sections field 3b is conspicuous by the presence of a thick and densely populated layer 4, whereas in field 3a this layer is present but is thinner and overall less defined. As the border of architectural field 4 is crossed, layer 4 disappears completely. There are large, conspicuous cells in layer 5 of field 3a but not in 3b (Fig. 1). Although the layer 5 pyramidal cells in field 3a are on average smaller than those found in field 4, there is sufficient overlap in the distribution of sizes to demand the use of additional criteria in defining the caudal border of the motor cortex. In myelin-stained sections throughout most of the mediolateral extent of M1, field 3a can be easily seen as a less myelinated "gap" separating fields 3b and 4 (Figs. 1, 2A).

The architectural characteristics of M1 change according to mediolateral level, making these distinctions less clear in the most lateral regions, from which facial movements are elicited (Figs. 2B–D, 3). "Giant" isolated pyramidal cells (Betz neurons) are conspicuous near the midline (Fig. 2B) but become gradually smaller in the forelimb representation (Figs. 1, 2C). Given the likely homology

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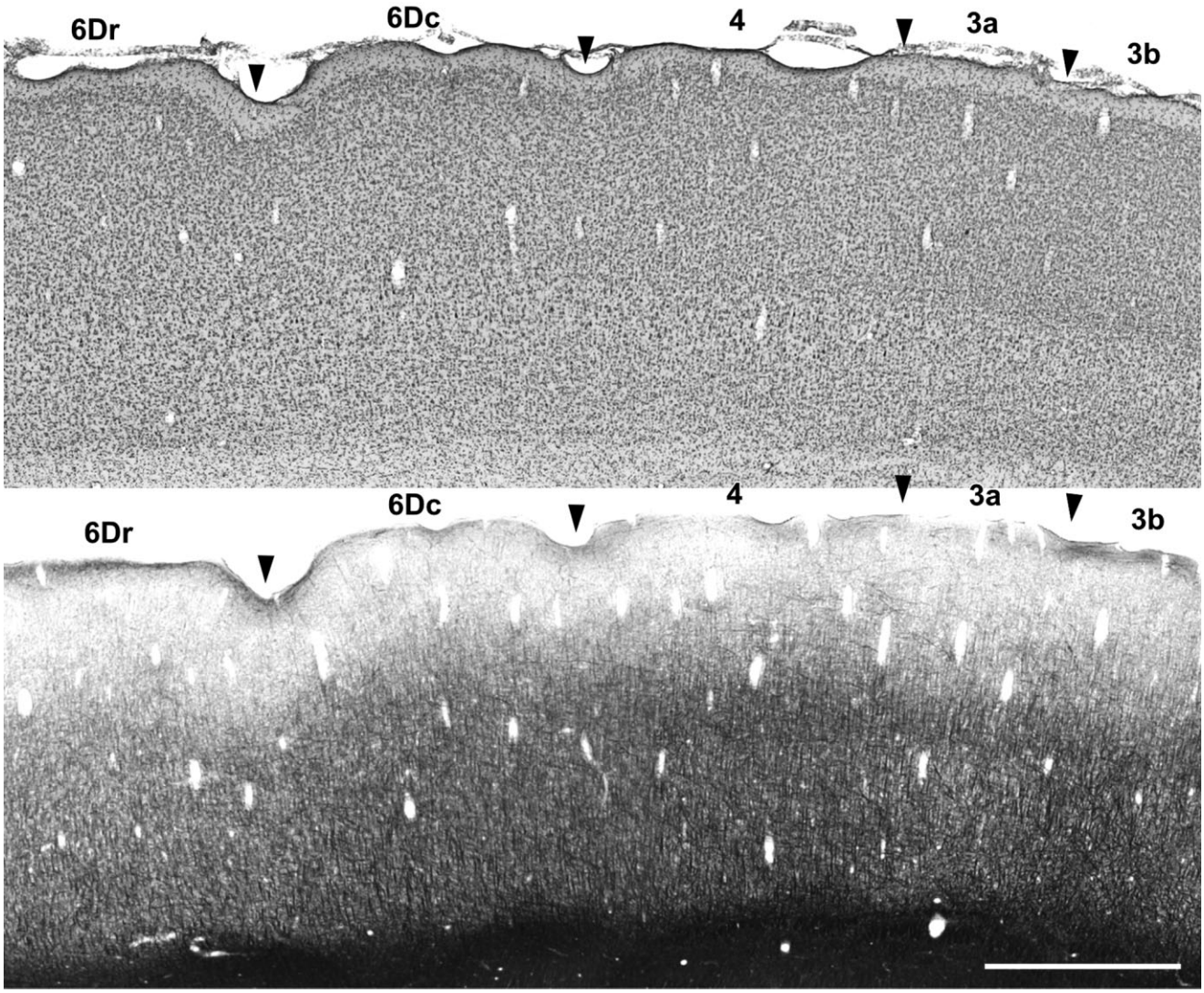


Fig. 1. Parasagittal sections through the rostral parietal and caudal frontal cortices of a marmoset (ML level approximately +5.0), illustrating the cytoarchitectural (top) and myeloarchitectural (bottom) characteristics of somatosensory (3a, 3b) and motor (4, 6Dc, 6Dr) fields. The bottom section was stained by using Schmued (1990) gold chloride stain; similar patterns were observed in Gallyas-stained materials. Note the sharply defined layer 4 and heavy myelination of field 3b, as well as the lower myelination and relatively large layer 5 cells in area 3a. Fields 4 and 6Dc both appear as densely myelinated, although the bands of Baillarger are more defined in field 4. The rostral border of field 6Dc is marked by a concomitant sudden reduction in the size of layer 5 pyramidal cells and myelination. Rostral is to the left. Scale bar = 1 mm.

with cytoarchitectural fields 4a and 4b of other primate species, and the gradient-like change in architectural features, we refer to the main portion of the marmoset M1 as field 4.

Furthermore, in agreement with studies in Old World monkeys, we found that in the lateral part of M1 the pyramidal cells in layer 5 are relatively small, being comparable in size to the analogous neurons in premotor areas, and that a thin layer 4 can be usually distinguished (Fig. 2D). In addition, the lateral part of M1 is slightly less myelinated than the medial part, while retaining the subtle separation between the inner and outer bands of Baillarger that distinguishes this area from the dorsal premotor cortex (Fig. 3; see also below). We adopted the designation field 4c to refer to this subregion of M1 (Vogt and Vogt, 1919; Barbas and Pandya, 1987; Watanabe-Sawaguchi et al., 1991). The exact boundaries between architectural fields are difficult to determine in this lateral region, resulting in relatively large regions of uncertainty, particularly in specimens sectioned in the parasagittal plane. In our experience, the myelin stains provide the best definition of the extent of M1.

Architectural fields along the rostral border of M1. Figure 1 shows that in the marmoset the cortex populated by large pyramidal cells in layer 5 is wide, extending to at least 3–4 mm anterior to field 3a. However, there are differences between caudal and rostral parts of this region: in the caudal region layer 5 is thicker, and the

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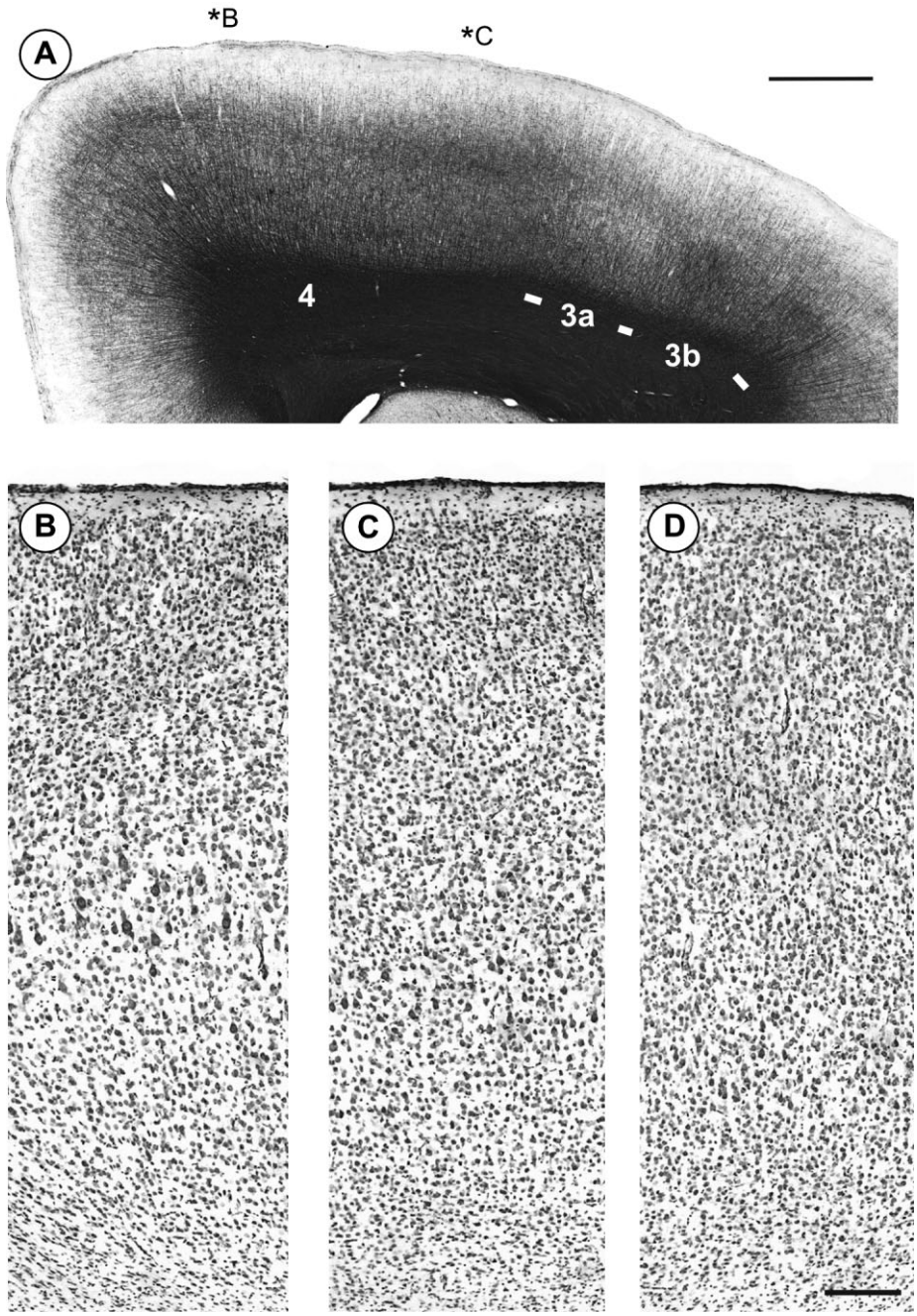


Fig. 2. **A:** Coronal section through the caudal portion of the frontal lobe of a marmoset (approximately at AP +9.5; medial to the left), stained for myelin using the Gallyas (1979) technique. Note the myelination "gap" corresponding to area 3a, which defines the ventral border of M1 at the level of representation of the axial and limb musculatures. **B–D:** Mediolateral differences in the cytoarchitecture of M1. B: Medial portion. C: Intermediate portion. D: Lateral portion (field 4c). The asterisks in A indicate the locations from which strips B and C were photographed, in the adjacent Nissl-stained coronal section. The location of strip D is indicated in a similar manner in Figure 3. Scale bar = 1 mm in A; 200 μm in D (applies to B–D).

pyramidal neurons on average larger, than in the rostral region. A correspondingly subtle transition can be observed in myelin-stained sections (Fig. 1): the caudal part tends to show an incipient separation between the inner and outer bands of Baillarger, whereas the rostral part shows little evidence of lamination. Based on the results of physiological and anatomical tracing experiments described below, we refer to the rostral subdivision as field 6Dc (Barbas and Pandya, 1987). As reported earlier (Burman et al., 2006) field 6Dc is separated from the granular frontal cortex (architectural fields 8B, 8Ad, and 8Av) by another architectural field (6Dr), which lacks the deeply

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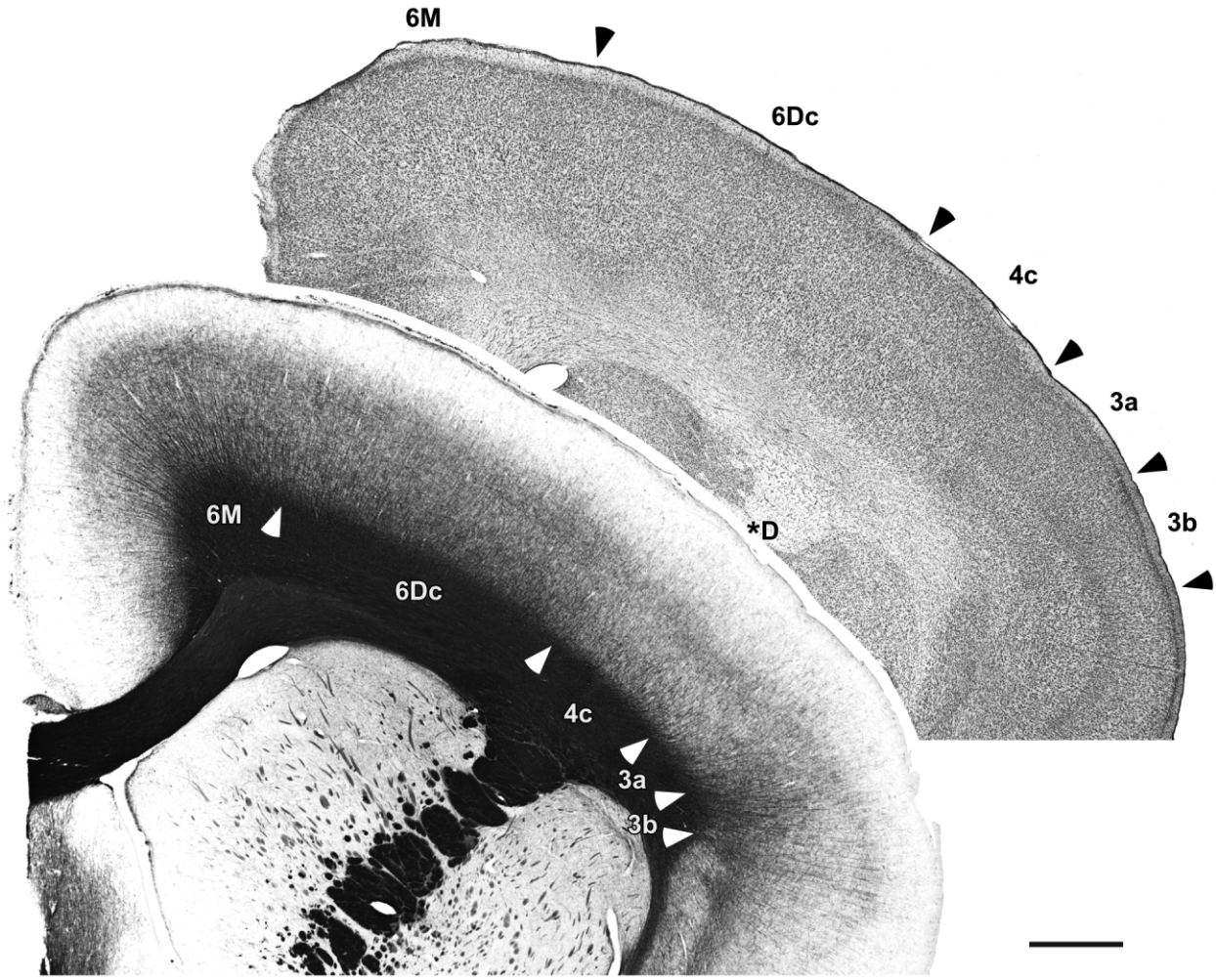


Fig. 3. Coronal sections approximately at AP +12.0, stained for myelin (Gallyas stain, foreground) and Nissl (background). This level cuts through the caudal premotor areas, the face representation in M1 (cytoarchitectural field 4c), and the rostral tips of areas 3a and 3b. Note that field 4c appears as less myelinated than field 6Dc but is more myelinated and homogeneous than field 3a. The asterisk adjacent to the myelin-stained section indicates the location of strip D illustrated in Figure 2. Medial is to the left. Scale bar = 1 mm.

stained, large layer 5 neurons that characterize both fields 4 and 6Dc in Nissl and cytochrome oxidase preparations (Figs. 1, 4). Field 6Dr is also distinct from 6Dc by virtue of staining more lightly for myelin, although it retains the characteristic homogeneous myelination from the white matter to the middle of layer 3 (Fig. 1).

Medial to 6Dc is another subdivision of the premotor cortex, which we refer to as field 6M. Limited microstimulation results (see below) indicate that the cortex in this region is electrically excitable, suggesting that it may include the supplementary motor area (Huffman and Krubitzer, 2001). In comparison with fields 4 and 6Dc, field 6M shows an incipient layer 4 in Nissl stains, as well as a greater separation between the inner and outer bands of Baillarger and conspicuous bundles of thick myelinated fibres radiating from the white matter, in myelin stains (Fig. 3).

Lateral to 6Dc we detected two architectural fields that have characteristics similar to those associated with areas 6Va and 6Vb in the macaque (Barbas and Pandya, 1987). However, with the exception of a few sites where high-threshold facial movements were elicited (see Fig. 7), our electrode penetrations failed to reach this region. Although the location of this putative ventral premotor cortex corresponds to that in other species of New World monkey (Dancause et al., 2006; Stepniewska et al., 2006), in the absence of physiological evidence we regard its identification in the marmoset (Fig. 5) as tentative. The dorsal part of this field (6Va) was more myelinated than its ventral part (6Vb).

Relationship between architectural fields and somatotopic representation

Figure 6 illustrates the relationship between the somatotopic representation of M1, revealed by intracortical microstimulation in three marmosets, and architectural fields. Jaw and facial movements were observed ventrally, largely within architectural field 4c. Most of these movements consisted of jaw opening and closing, although nose, lip, and eyebrow twitches were also observed in a few

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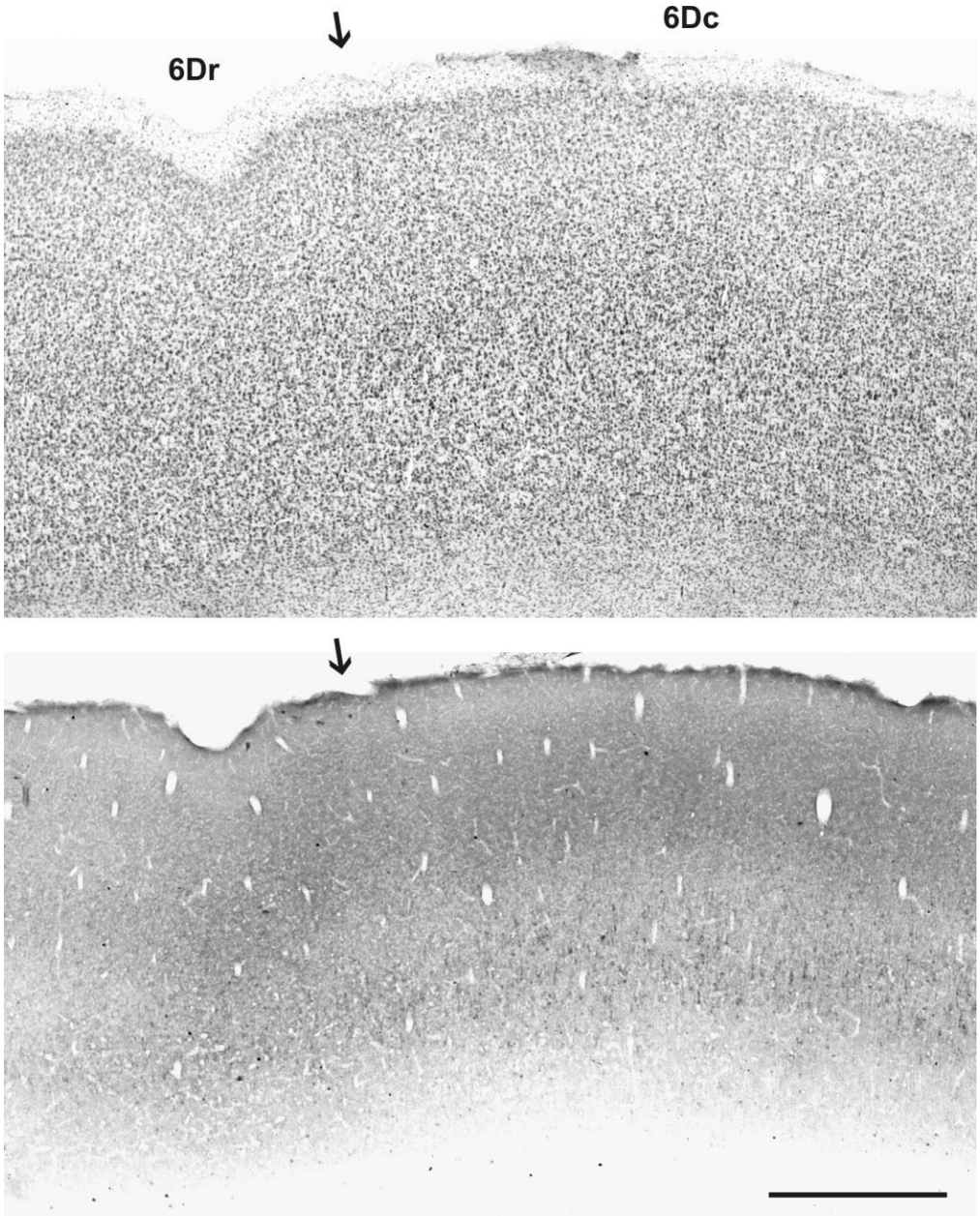


Fig. 4. Parasagittal sections (ML +2.8) stained for Nissl (top) and cytochrome oxidase (bottom), illustrating the boundary between premotor fields 6Dc and 6Dr. Note that the reduction in size of layer 5 pyramidal cells that is visible in the Nissl stain is paralleled by a reduction in the intensity of cytochrome oxidase staining (in particular, the densely stained cells in layer 5). Rostral is to the left. Scale bar = 1 mm.

sites. In contrast, with the possible exception of sites near architectural boundaries, stimulation of sites within field 4 only elicited movements of the hindlimb (HL), axial (A), and forelimb (FL) musculatures, in mediolateral sequence. Relatively large territories were occupied by the representations of the forelimb and face (in particular, movements consisting of simultaneous flexion of the fingers, wrist flexion, and jaw opening). Although sites eliciting movements of distal and proximal forelimb tended to cluster (indicated by "p" and "d" in Fig. 6), in at least one case two regions of proximal representation were separated by a large expanse of distal (wrist and fingers) representation, suggesting that a mosaic-like pattern may exist at a finer scale (Gould et al., 1986). The mediolateral sequence of the somatotopic map was similar to that reported in other primate species, as suggested by an earlier brief report (Woolsey et al., 1949).

Reflecting earlier observations in the oculomotor cortex (Blum et al., 1982), we found that the generation of overt movements in the marmoset required relatively high microstimulation currents in comparison with those reported in larger primate species (Fig. 7). However, differences in

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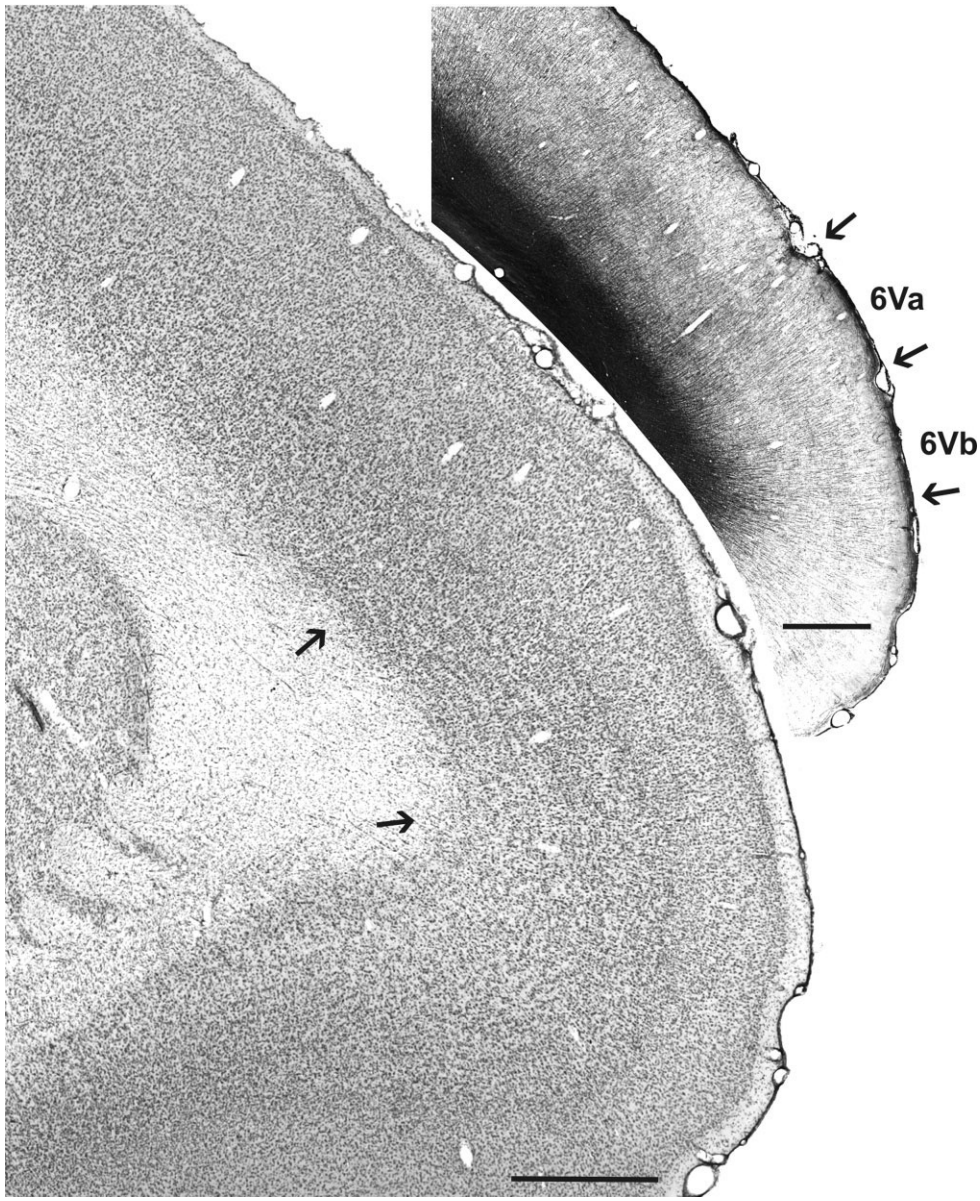


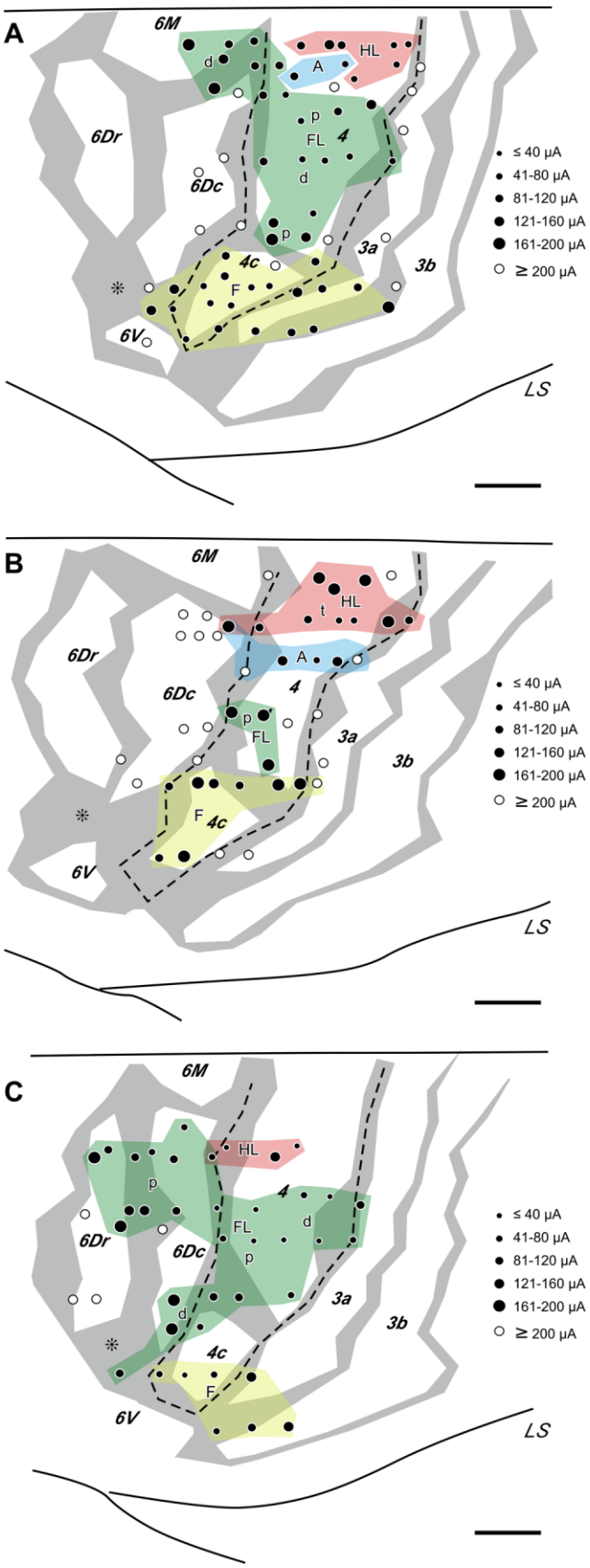
Fig. 5. Definition of a putative homologue of area 6V in the marmoset, illustrated in coronal sections through the lateral frontal cortex (approximately AP +13.0). Field 6V, defined on the Nissl-stained material (foreground) appears to encompass dorsal and ventral subdivisions, based on the density of myelination (6Va and 6Vb; Barbas and Pandya, 1987). Immediately dorsal to 6Va there is cortex likely to include the posterior tip of the frontal eye field (area 8Av; see Burman et al., 2006) and premotor field 6Dr. Scale bar = 1 mm.

excitability were always observed upon crossing the boundaries between M1 (architectural fields 4 and 4c) and the rostral fields. In two of the animals (Fig. 6A,B) sites in area 6Dc were unresponsive to currents below 200 μA, whereas high-threshold stimulation (>120 μA) of a few sites rostral to the medial and lateral extremes of M1 (putative areas 6M and 6Va) elicited movements of the forelimb and mouth, respectively. In another animal (Fig. 6C), motor responses could be elicited from fields 6Dc and 6Dr, albeit at much higher thresholds in comparison with those observed in fields 4 and 4c. Most of the movements elicited from these fields consisted of arm, wrist, and finger flexions. Within M1, the lowest microstimulation thresholds corresponded to the representations of the hand and wrist (Fig. 7), reflecting earlier reports in other species (Wong et al., 1978; Sessle and Weisendanger, 1982).

Anatomical confirmation of the border between M1 and the premotor belt

Reports in other species of primate have indicated that posterior parietal connections of M1 focus on the rostral portions of this region (area 5), which are primarily related to processing of somatosensory inputs (e.g., Leichnetz, 1986; Darian-Smith et al., 1993; Fang et al., 2005). In contrast, tracer injections in the premotor areas often

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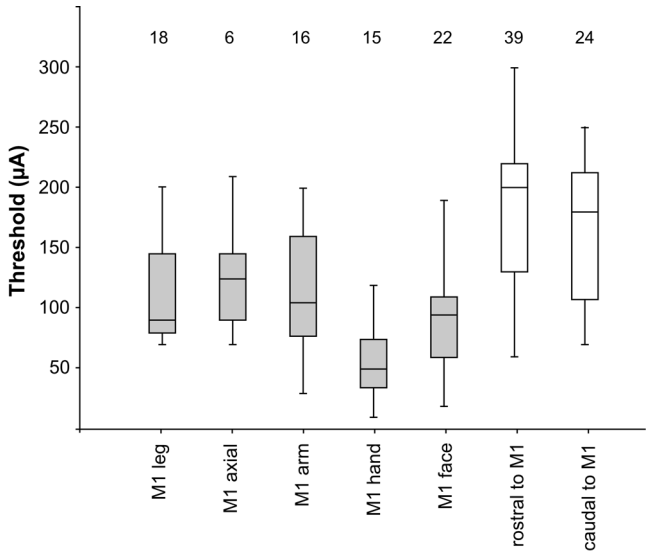


Fig. 7. Summary of the microstimulation thresholds observed in different parts of M1 (gray) and in adjacent areas (white), represented by Tukey box plots. Each box encompasses the second and third quartiles of the distributions, with the median values indicated by the horizontal lines. The "whiskers" indicate the lowest and highest values observed. The numbers above the figure indicate the number of sites obtained in each subdivision.

demonstrate additional connections with caudal regions of the parietal cortex, including high-order stations of the visual "dorsal stream" (e.g., Matelli et al., 1998; Marconi et al., 2001; Tanné-Gariépy et al., 2002; Rozzi et al., 2006). Thus, in order to help determine the rostral border of M1, we considered the pattern of projections between frontal fields and the posterior parietal cortex.

Four injections were placed in the caudal parietal lobe, including the medial posterior parietal region (PPm) and the cortex immediately rostral to it (PEc), where visual responses have been recorded (Rosa and Schmid, 1995; Rosa and Tweedale, 2005). These regions of the marmoset cortex are likely to correspond to areas V6A and PEc of Old World monkeys (Rosa and Tweedale, 2001; Galletti et al., 2003; Breveglieri et al., 2007). The mediolateral location of these injections varied between cases, including both the dorsal exposed surface of the parietal lobe (two

Fig. 6. Relationship between architectural fields in the posterior part of the frontal lobe and the somatotopic organization of the motor cortex, revealed by intracortical microstimulation. **A–C:** Dorsal views of the brains of three marmosets (CJ64, CJ60, and CJ61), showing the entry points of electrode penetrations (circles) relative to various architectural fields. Rostral is to the left and medial to the top. Electrode penetrations that yielded movements of the hindlimb (HL); trunk, neck, and tail (axial musculature, A); forelimb (FL); and jaw/face (F) are joined into red, blue, green, and yellow polygons, respectively. The sites where movements required currents at greater than 200 µA are indicated by the white circles. Gray zones correspond to zones of uncertainty (see Materials and Methods), and the dashed line is our best architectural estimate of the outer boundary of area M1. The asterisk indicates a transitional zone between areas 6 and 8, which is likely to be part of field 8Av (frontal eye field; see Burman et al., 2006). LS, lateral sulcus (sylvian fissure); d, distal; p, proximal. Scale bar = 1 µm in A–C.

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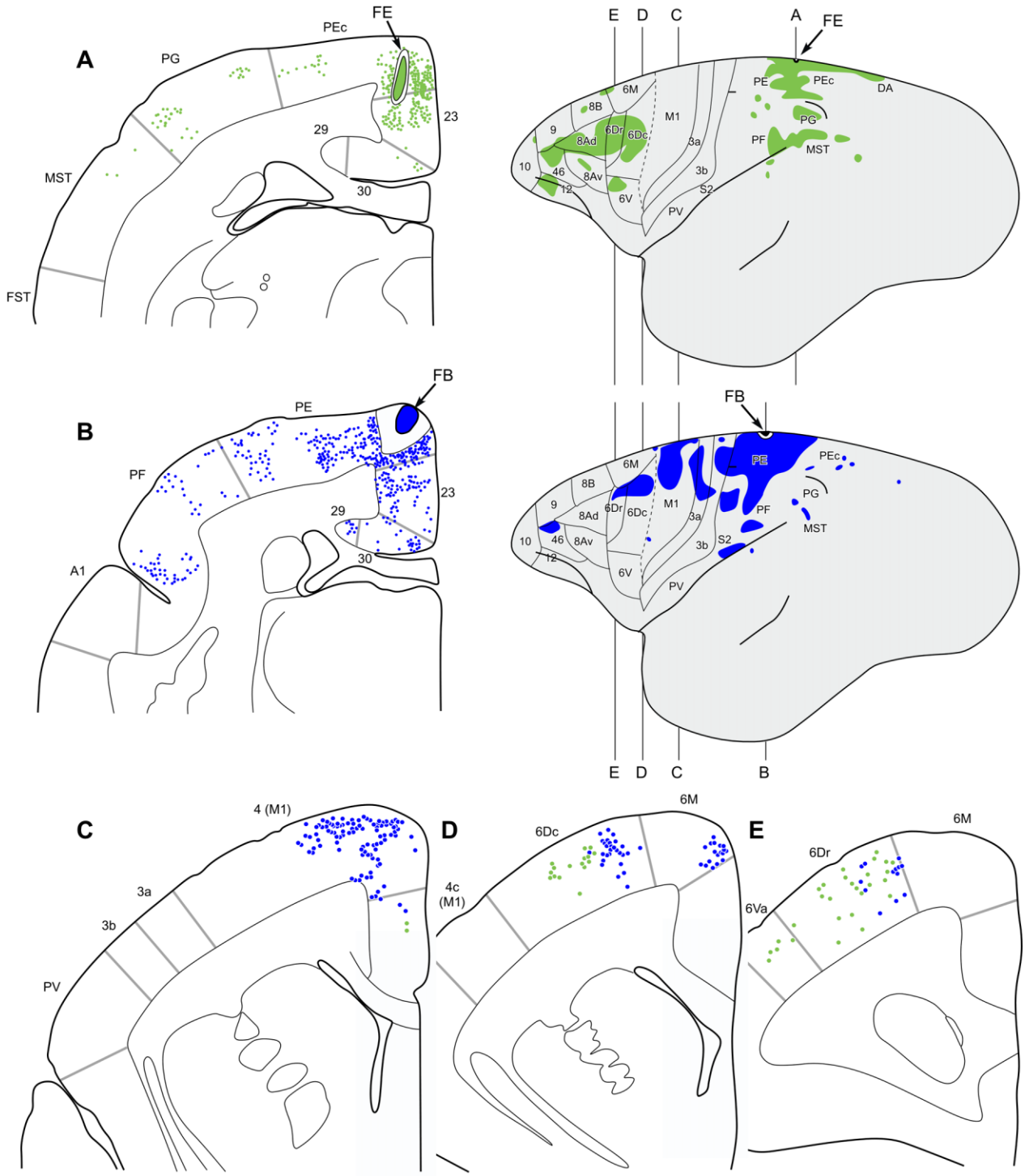


Fig. 8. Patterns of connections revealed by two tracer injections in the parietal lobe of a marmoset (CJ45L). Top and middle right: lateral reconstructions of the left hemisphere illustrating the location of patches of labeled cells relative to cortical areas (Krubitzer and Kaas, 1990; Rosa and Elston, 1998; Burman et al., 2006). The caudal injection (Fluoro Emerald [FE]; shown in green) was centered in a predominantly visual part of the posterior parietal cortex (cytoarchitectural field PEc), as evidenced by the labeled cells in extrastriate areas (label in the dorsoanterior [DA] and medial superior temporal [MST] areas is evident in these lateral views), and by the lack of label in somatosensory areas. The rostral injection (fast blue [FB]; shown in blue) crossed the border into a predominantly somatosensory subdivision of the posterior parietal cortex, as evidenced by the label in areas 3a, 3b, and S2. **A:** Coronal section illustrating the FE injection site and labeled connections. **B:** Coronal section illustrating the FB injection site and labeled connections. **C–E:** Coronal sections through the frontal lobe illustrating the location of labeled cells (FB, blue; FE, green). Note that the two injections labelled cells in fields 6Dc and 6Dr, whereas only FB-labeled cells were observed in M1 (fields 4 and 4c).

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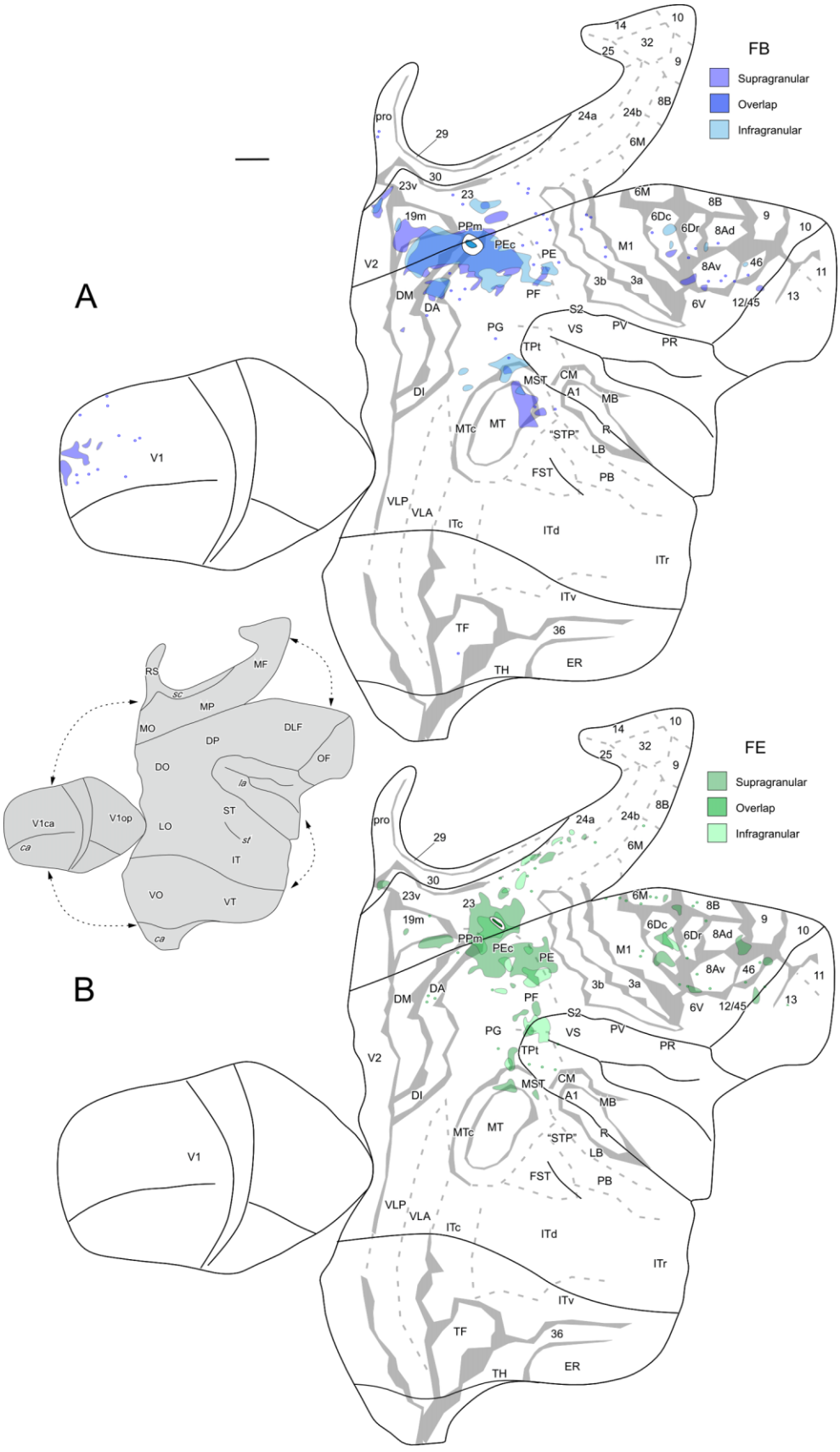


Figure 9 (continued)

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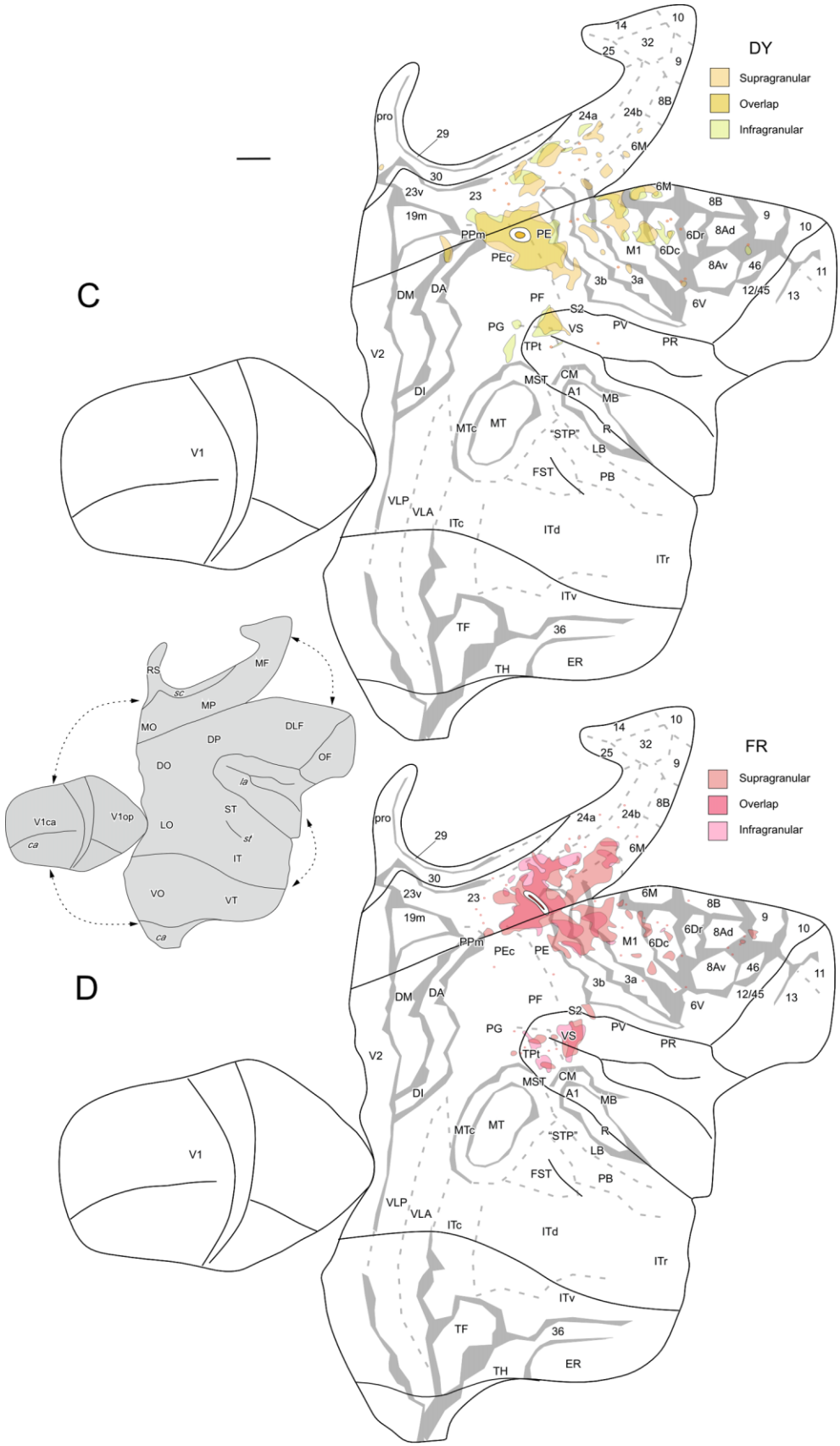


Figure 9 (continued)

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cases) and the cortex parallel to the midline (two cases). For comparison, three additional injections (two in the lateral surface, one along the midline) were located more rostrally, in cortex likely to correspond to the somatosensory association cortex. This region of cortex shows no evidence of subdivisions in the marmoset (Padberg et al., 2007; see also Carlson et al., 1986, for results in the closely related *Saguinus*) and thus for the purposes of the present analysis can be reasonably regarded as a single region (here designated as PE) that amalgamates fields 1, 2, and 5 of other species.

The primary finding from these cases is illustrated by the example shown in Figure 8: whereas fields 6Dc and 6M had substantial projections to the visual areas of the posterior part of the parietal lobe and occipitoparietal transition, area M1 had none. Instead, similar to other species of monkey, M1 only showed connections with the somatosensory parietal areas. Data from a second animal, in which a rostrocaudal progression of four parietal injections was performed, are illustrated in Figure 9. Supporting the results shown in Figure 8, the two caudal injections labelled cells in 6Dc but not in M1. These same injections resulted in retrograde label across many "dorsal stream" visual areas, including the dorsomedial (DM) and dorsoanterior (DA) areas, medial areas 19m and 23v, and the middle temporal (MT) and medial superior temporal (MST) areas (Palmer and Rosa 2006a,b; Fig. 9A,B). Further confirming its placement in visual cortex, the most caudal injection (Fig. 9A) even labeled the representations of the far peripheral visual field in V1 and V2, possibly due to touching the border of area DM (Rosa et al., 2005). In contrast, the rostral injections, which labeled projections from M1, resulted in label in somatosensory areas (3a, 3b, and the dorsal bank of the lateral fissure; Krubitzer and Kaas, 1990; Fig. 9C,D). In summary, these connectional results highlight the distinction between fields 4 and 6Dc and support the point of view that the latter is best considered a subdivision of the premotor complex.

DISCUSSION

Heterogeneity of M1

The key motivation behind the present study was to define the extent of the primary motor cortex (M1) in the marmoset, a small New World monkey. Earlier cytoarchitectural analyses of the frontal cortex in this species resulted in somewhat different views on the likely extent of the motor cortex. Brodmann (1909) recognized a very expansive area 4, which encompassed the entire region showing large pyramidal cells in layer 5 (i.e., not only M1, as defined here, but also most of the premotor cortex). In contrast, Peden and Von Bonin (1947) illustrated a much smaller motor area (field FA), followed by a premotor belt (FB) within the same region of cortex. Our preliminary assessment of the extent of the motor cortex in the marmoset (Burman et al., 2006) was consistent with Peden and Von Bonin's map. The present results require a minor revision of this point of view, as they indicate that M1 is likely to extend further, by including a ventral region adjacent to the rostral extension of areas 3a and 3b (Fig. 3). This region, which includes representations of face, jaw, and mouth movements, was previously thought to be part of the ventral premotor cortex. The present architectural analyses (for summary, see Fig. 10) suggest that area 6V is in fact located further rostrally, and slightly ventrally in relation to the extended M1.

The architectural heterogeneity of M1 is consistent with some of the earlier reports in Old World monkeys (in particular, Vogt and Vogt, 1919; Barbas and Pandya, 1987; Watanabe-Sawaguchi et al., 1991). The recognition that M1 is architecturally heterogeneous indicates parallels with the premotor belt, where the dorsal and ventral subdivisions of area 6 are known to emphasize movements of different body parts and to have different architectural patterns (e.g., Barbas and Pandya, 1987; Stepniewska et al., 1993). Similar to M1, the dorsal portions of the premotor belt also contain larger pyramidal cells in layer 5. As pointed out by Graziano and Aflalo (2007), it is sometimes difficult to define exactly what is a "cortical area" in

Fig. 9. Patterns of connections revealed by four tracer injections in the parietal lobe of a marmoset (CJ58R). **A,B:** Injections of fast blue (FB, blue) and Fluoro Emerald (FE, green) in caudal regions (PPm and PEc), which are related to vision. **C,D:** Injections of diamidino yellow (DY, yellow) and Fluoro Ruby (FR, red) in rostral regions, which are predominantly related to somatic sensation. Each diagram is a graphic reconstruction of the "unfolded" cerebral cortex, prepared in the style suggested by Van Essen and Maunsell (1980; see also Palmer and Rosa, 2006a). Patches of label in the supragranular and infragranular layers are indicated by different tones and single cells by the small circles. Gray zones represent boundary regions between architectural fields, and dashed gray lines represent borders that were estimated based on results in other animals. Continuous lines represent the boundaries of the maps as well as the main folds of the cerebral cortex. **Insets (left):** Gross anatomical features of the cerebral cortex as they appear in this format, as well as the discontinuities introduced in the map to allow minimal distortion (arrows). Abbreviations (insets): ca, calcarine sulcus; DLF, dorsolateral frontal cortex; DO, dorsal occipital cortex; DP, dorsal parietal cortex; IT, inferior temporal cortex; la, lateral sulcus; LO, lateral occipital cortex; MF, medial frontal cortex; MO, medial occipital cortex; MP, medial parietal cortex; OF, orbitofrontal cortex; RS, retrosplenial cortex; sc, supracallosal sulcus; st, superior temporal sulcus; ST, superior temporal cortex; V1ca, primary visual area, calcarine region; V1op, primary visual area, opercular region; VO, ventral occipital cortex; VT, ventral temporal cortex. Abbreviations (main panels): numerical designations correspond to architectural fields defined in the marmoset by Krubitzer and Kaas (1990; fields 3a and 3b), Burman et al. (2006; fields 6Dc, 6Dr, 6M, 6V, 8Ad, 8Av, 8B, 9, 10, 11, 13, 12/45, 46), and Palmer and Rosa (2006a,b; fields 19m, 23, 23v, 29, 30 and 36). Other designations: A1, primary auditory area; CM, caudomedial area; ER, entorhinal cortex; FST, fundus of superior temporal area; ITc, caudal subdivision of the inferior temporal cortex; ITd, dorsal subdivision of the inferior temporal cortex; ITr, rostral subdivision of the inferior temporal cortex; ITv, ventral subdivision of the inferior temporal cortex; LB, lateral auditory belt; M1, primary motor area; MB, medial auditory belt; MST, medial superior temporal area; MT, middle temporal area; MTc, middle temporal crescent; PB, auditory parabelt; PE, parietal field E (Peden and Von Bonin, 1947); PEc, caudal subdivision of parietal field E; PF, parietal field F; PG, parietal field G; PPm, medial subdivision of the posterior parietal cortex; PR, parietal rostral area; PV, parietal ventral area; R, rostral auditory area; S2, second somatosensory area; STP, superior temporal polysensory area; TF, temporal field F; TH, temporal field H; TPt, temporoparietal transitional field; V1, primary visual area; V2, second visual area; VLA, ventrolateral area; VLP, ventrolateral posterior area; VS, ventral somatosensory area.

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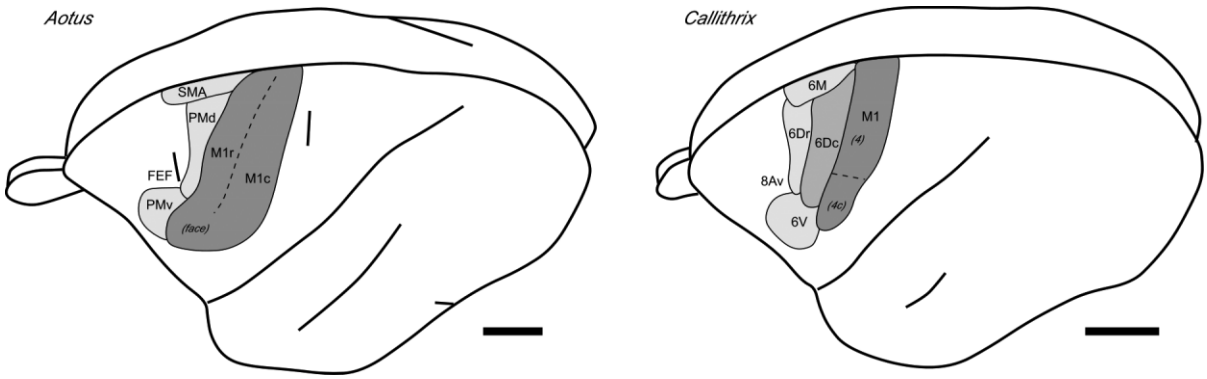


Fig. 10. Comparison of the subdivisions of the caudal frontal cortex proposed for the owl monkey (left; based on Stepniewska et al., 1993) and marmoset (present results). The light gray indicates regions that are considered part of the premotor cortex in both species. The rostral sudivision of M1 of the owl monkey (M1r) is likely to overlap with the caudal subdivision of the dorsal premotor cortex (field 6Dc) in the marmoset. Scale bar = 5 mm.

the frontal lobe, given that boundaries appear to be intrinsically criterion dependent. In the present case, if one assumes that M1 contains a complete body representation, then it must also be accepted that this area is not a single architectural entity. Similar arguments have been made with respect to the definition of subdivisions of the association cortices of the temporal (Rosa and Tweedale, 2005; Palmer and Rosa, 2006b) and parietal (Breveglieri et al., 2007) lobes, where different parts of a same area (defined by topographic representation and/or architecture) have been shown to differ in terms of connections and response properties. It is also the case that architectural patterns in the frontal lobe have been shown to undergo subtle changes as a function of age, even in postnatal life (Burman et al., 2007), thus providing a reasonable mechanism to explain slight individual differences in histological appearance by differential preservation/elimination of cell populations and connections. Given these caveats, a histological definition of "what M1 is" is best expressed in relation to adjacent areas, rather than in absolute terms.

Rostral boundary of M1 with the premotor cortex

Earlier work in another species of New World monkey, the owl monkey, has reported the existence of a large M1 consisting of two subdivisions in rostrocaudal succession (M1r and M1c; Stepniewska et al., 1993), followed by a relatively narrow premotor belt. Based on cytoarchitectural characteristics, it is likely that M1, as defined in the present study, corresponds to M1c, whereas field 6Dc corresponds to M1r. Indeed, examination of a parasagittal section illustrated by Stepniewska et al. (1993; their Figure 4) suggests that the dorsal premotor area (PMd) defined in the owl monkey has the architectural characteristics of area 6Dr of the nomenclature we favor (based on Barbas and Pandya, 1987; see also Wise, 1985). Likewise, Huffman and Krubitzer (2001) illustrate a very wide M1 in the marmoset, based on dense myelination observed in sections tangent to the surface of the brain. In contrast, the present results indicate that dense myelination is present both in field 4 (part of M1) and in field 6Dc, which nonetheless show subtle differences in the separation between bands of Baillarger when they are studied in cross section (see Fig. 1). While recognizing the architectural similarities between areas 4 and 6Dc, we believe that the present nomenclature is appropriate to describe the organization of the marmoset cortex, given its pattern of corticocortical connections and degree of electrical excitability. In particular, the connections with the caudal part of the parietal cortex (regions likely to correspond to areas V6A, MIP, and PEc of the macaque; Rosa and Tweedale, 2001; Galletti et al., 2003) are likely to reflect a role of area 6Dc in visuomotor integration and planning (Preuss et al., 1996; Burman et al., 2006; Breveglieri et al., 2007), as opposed to movement execution.

Despite the different nomenclatures, there is agreement across studies on different species on the existence of two functional subdivisions (M1c and M1r, or 4 and 6Dc) within the heavily myelinated frontal region containing large pyramidal cells in layer 5 (Fig. 10; see also Wise, 1985). Fields 6Dc and 6Dr are likely to correspond to areas F2 and F7, respectively, according to the nomenclature proposed by Matelli et al. (1985) for the macaque premotor cortex.

Injections of tracers in the owl monkey's M1r are reported to result in denser label in posterior parietal areas, compared with injections in M1c (Stepniewska et al., 1993). However, no detail is available regarding the specific parietal areas involved in these connections, so it remains unclear whether visual areas project to M1r, as it would be expected if this subdivision corresponded to the marmoset's 6Dc. In the prosimian *Galago*, in which caudal and rostral subdivisions of the primary motor cortex were not recognized, injections in the parietal visual cortex result in labeled cells just rostral to the border of M1 (Fang et al., 2005). These results resemble the present observations in the marmoset, as well as the results of earlier work in the macaque (Matelli et al., 1998; Shipp et al., 1998; Caminiti et al., 1999; Lewis and Van Essen, 2000; Tanné-Gariépy et al., 2002).

Representation of orofacial movements in the marmoset

Our results support those of Stepniewska et al. (1993) in suggesting that the representation of the lower face and jaw in M1 extends ventrally as far as the rostral tip of somatosensory area 3a. In recent years the marmoset has

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become a choice primate model for studies of intraspecific vocal communication (Wang et al., 1995; Eliades and Wang, 2003, 2005; Cheung et al., 2005). Although much work has been done in terms of understanding the analysis of vocalization calls in the auditory cortex, relatively little is known about the cortical areas involved in the production of mouth and tongue movements. In this context, the knowledge of the relationship between histology and functional topography obtained in the present study is likely to become relevant for future work on the neural circuitry involved in the production of vocalizations, including connections, neurophysiology, or noninvasive imaging data. Movements of the face, including predominantly those of the mouth, were observed after the stimulation of sites that collectively corresponded to nearly one-quarter of the extent of M1. The cortical region capable of eliciting orofacial movements in the marmoset appears different from that reported in the closely related tamarin monkey, in a study that used high-stimulation currents to guide tracer injections to the motorcortical tongue area (Alipour et al., 2002). According to the current understanding of the cerebral cortical organization in New World monkeys, the tracer injection site illustrated in this earlier study appears to be located in one of the ventral somatosensory fields, which are likely to subserve gustatory functions also (Kaas, 2005). Nonetheless, the injection illustrated by Alipour et al. (2002) resulted in dense anterograde label in the expected location of cytoarchitectural field 4c (their Fig. 1b, stereotaxic levels AP +11.5 and +12.5).

Excitability of the marmoset motor cortex

The minimum level of current required to elicit a response within the marmoset primary motor cortex was relatively high in comparison with other primate species studied to date. For example, Gould et al. (1986), Stepniewska et al. (1993), and Preuss et al. (1996) reported microstimulation thresholds of less than 30 μA to be relatively common in the caudal part of owl monkey M1 (M1c), an observation that reflects earlier findings in the macaque (e.g., Sessle and Wiesendanger, 1982). Although a few such sites were observed in the present experiments (Fig. 7), in most cases substantially higher currents (>60 μA) were required. Similarly, a marked difference has been previously described with respect to the excitability of the marmoset and squirrel monkey frontal eye fields (Blum et al., 1982), and Burish et al. (2006) reported that elicitation of movements in the premotor areas of the marmoset require high-threshold stimuli. Indeed, it is possible that some of the movements we obtained using currents of 200 μA or above were due to current spread to sites in M1. As a control for technical issues, we have performed experiments in rats using the same protocol used in the marmosets, in which we were able to drive overt movements using currents similar to those reported in the literature (Donoghue and Wise, 1982; Neafsey et al., 1986). Thus, there are good reasons to suspect that genuine interspecies differences exist in this respect.

Although it is possible that differential sensitivity to anesthesia may be the reason for the observed differences between primate species, other factors may also play a role. Corticospinal projections in the marmoset reach the medial and dorsal regions of the ventral horn but do not reach the vicinity of the motoneuron pools (Galea, 1997). Moreover, even the largest pyramidal cells in layer 5 of the marmoset M1 are relatively small, in comparison with those found in the other primate species (Nudo et al., 1995) and thus are likely to give rise to thinner axons (e.g., Clamann and Henneman, 1976; Walsh et al., 1999). Constraints related to the summation of corticospinal inputs through an indirect pathway (Bortoff and Strick, 1993; Nakajima et al., 2000) as well as a greater input resistance of descending axons to electrical stimulation (Jack et al., 1983; Rubinstein, 1991; Nowak and Bullier, 1998) may thus converge to result in a relative lack of excitability, in comparison with larger primates.

Comparative aspects of frontal cortical organization

The marmoset brain has less than one-twelfth of the mass of the macaque brain, which is in turn even smaller relative to the mass of the human brain (Stephan et al., 1981). Knowledge of the cortical organization in the small brain of the marmoset holds considerable interest in terms of understanding aspects of brain scaling within the order primates, including issues such as the remodeling of areal boundaries and connections as a function of size (Carlson et al., 1986; Changizi and Shimojo, 2005; Palmer and Rosa, 2006a; Preuss and Goldman-Rakic, 1991). Although the present results demonstrate that the caudal portion of the frontal lobe of the marmoset contains the same basic subdivisions found in the much larger macaque brain, they also reveal differences. The motor and premotor cortices form a relatively large proportion of the frontal lobe, at the apparent expense of the granular frontal areas (Figs. 8, 9). Given that the postnatal maturation of the frontal cortex occurs along a caudal to rostral gradient, starting from the motor area (Burman et al., 2007), this finding is compatible with other evidence suggesting that early-maturing neural structures become disproportionately large in smaller species, which have shorter developmental periods (Finlay and Darlington, 1995; Rosa, 2002). In addition, similar to our earlier findings about the granular frontal cortex (Burman et al., 2006), some of the finer histological subdivisions that are apparent in larger simian brains do not appear obvious in the marmoset (for example, the additional subdivisions of agranular cortex that exist along the rostral boundaries of fields 4 and 6 in baboons; Watanabe-Sawaguchi et al., 1991).

This provides further evidence of addition and differentiation of cortical fields as a mechanism for cortical expansion in larger primate species (Ebbesson, 1984; Carlson et al., 1986; Krubitzer, 1995; Padberg et al., 2007). Thus, the organization of the cerebral cortex in marmosets provides a good model for furthering our understanding of the developmental and evolutionary mechanisms responsible for the diversity of primate brains, while at the same time presenting us with the components of a simian cortex in their simplest expression.

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